

Praneeth Reddy Penamalli

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EDUCATION

Master of Science, Computer Science

The University of Texas at Dallas

Aug 2023 - Aug 2025

Relevant Coursework: Data Structures & Algorithms, Database Design, Artificial Intelligence, Big Data Management & Analysis, Web Programming Languages, Machine Learning, Natural Language Processing, Design & Analysis of Algorithms

Bachelor of Technology, Mechanical Engineering

Indian Institute of Information Technology Jabalpur

Aug 2017 - Jun 2021

TECHNICAL SKILLS

Languages & Scripting: C, C#, Java, Python, Ruby, TypeScript, SWI-Prolog, HTML5, CSS3, JavaScript, XML, JSON, PL/SQL

Frameworks & Tools: NumPy, Pandas, Keras, Scikit-learn, OpenCV, Dlib, PySpark, Apache Hadoop, MongoDB, React.js, Node.js, .NET Core, Flask, Angular, Selenium, Kafka, Spring Boot, RESTful APIs, Bootstrap, NLTK, Express.js, ELK Stack

Development & DevOps: Git, AWS, Docker, Azure, DevOps, Maven, CI/CD Pipeline, SDLC, Agile, Waterfall, Figma, JIRA

PROFESSIONAL EXPERIENCE

Accenture Solutions, Bangalore, India

Jan 2022 - Jul 2023

Advanced Application Engineering Analyst

- Pioneered an advanced Entity-Relationship model design that enhanced data retrieval efficiency by 20%.
- Revamped PostgreSQL operations with advanced normalization strategies, intelligent indexing, and streamlined DML and DDL, achieving a 17% system performance increase, enhancing database efficiency and responsiveness.
- Optimized critical web application performance by refactoring complex SQL queries and implementing advanced indexing strategies, resulting in a 25% improvement in page load times and reduced server resource load.
- Collaborated with a 20-member Agile development team to conduct Azure DevOps code reviews, ensuring code quality, adherence to best practices, and seamless integration for continuous delivery pipeline processes.
- Designed comprehensive end-to-end test automation using Selenium WebDriver with a Behavior-Driven Development (BDD) approach, validating functionality, regression, and performance for optimal user experience.
- Engineered ETL pipelines for user data extraction, leveraging PowerBI for analytics and strategic dashboards.
- Developed reports for data-driven decision-making, implementing validation protocols to ensure accuracy.

Cognizant Technology Solutions, Hyderabad, India

Sep 2021 - Jan 2022

Programmer Analyst

- Contributed to a hands-on frontend web development training project using C#, HTML, CSS, and JS technologies.
- Developed user-centric interfaces by implementing core UX principles for improved usability and engagement.

PROJECT EXPERIENCE

AI-Powered SQL Injection Defense System

Aug 2024 - Dec 2024

- Fine-tuned a DistilBERT model for real-time SQLi detection, enabling robust malicious input recognition.
- Architected a full-stack PHP/MySQL application with a vulnerable backend to demonstrate attack surfaces.
- Integrated the ML inference pipeline into the backend, enabling transparent blocking of malicious queries.

Autism Detection in Toddlers

Jan 2021 - May 2021

- Engineered a toddler eye gaze detection system with three monitors, using OpenCV and dlib libraries.
- Utilized dlib's 68 facial landmark detectors to accurately estimate the patient's gaze within a simple system.
- Utilized collected data to develop a robust classification model aimed at early detection of autism in toddlers, effectively reducing both time and costs traditionally associated with autism diagnosis procedures.

Article Recommendations

Jul 2020 - Nov 2020

- Engineered a TuriCreate based machine learning model utilizing TF-IDF to recommend pertinent articles.
- Explored document representations like bag-of-words and word embeddings for effective text processing.
- Implemented a highly effective text-based search algorithm utilizing cosine similarity and expertly leveraged structural document topics to significantly enhance recommendation precision for optimized content delivery.